

strataq: quantal response equilibria, their derivatives, their geometry and their dissipation, in one JAX library

Sharath Sathish*

August 2026 — software paper, draft v0.3

Abstract

strataq is an open-source JAX library for logit quantal response equilibrium (QRE) analysis and for three things routinely needed *after* an equilibrium that no existing package provides: the exact derivative of the equilibrium with respect to payoffs (the strategic resolvent $\chi^{\text{eq}} = (I - SB)^{-1}S$, which no solver exposes); the Candogan–Menache–Ozdaglar–Parrilo potential/harmonic decomposition, of which no implementation exists on any package index fifteen years after publication; and the non-equilibrium thermodynamics of the revision dynamics — currents and entropy production of the joint-profile Glauber chain — which existing irreversibility estimators cannot compute because they take a scalar series and have no concept of an agent. A fourth gap is closed by a packaged estimation workflow with named intervals, documented as missing by the reference solver’s own maintainers. On top sits a problem-oriented API (pricing, auction, routing, allocation, electricity, repeated, evolutionary and extensive-form problems), a one-call `diagnose()` whose refusals are returned as bounds, and publication figures on a shared palette. The paper’s argument is its calibration table: fourteen readings on systems whose correct answer is fixed by an analytic identity or an external oracle, each mapped to a committed artifact that regenerates from a recorded seed. The reciprocity defect reads $\leq 8.9 \times 10^{-17}$ on exact potential games and 5.6×10^{-17} on a real road network; the equilibrium derivative matches central finite differences to 1.3×10^{-8} and an independent implicit-differentiation path to 5.6×10^{-17} ; fixed- λ profiles match `pygambit`’s homotopy to 1.2×10^{-9} , and the agent-QRE solver matches `pygambit`’s extensive-form solver on four classic trees inside the library’s 10^{-8} oracle tolerance. We describe the evidence discipline the project runs on, including the release-integrity failure (F-0018) in which 23 in-repo gates were green while every solver call in the shipped wheel raised `FileNotFoundError`, and the tests that exist because of it. Limitations are stated in the same voice: the exact dynamics layer is dense and small-game only, the product surface above the gated instrument layer is tested but not yet gated, the extensive-form oracle checks are skipable, and the public API is broader than `__all__` documents.

Keywords: quantal response equilibrium; logit; comparative statics; potential games; Hodge decomposition; entropy production; JAX; research software

*With agent-assisted implementation. Source, gate records and every number’s regeneration path: <https://github.com/SharathSPhD/sage>. Correspondence: sharath.ai.colab@gmail.com.

1 Statement of need

Quantal response equilibrium [1] is the standard structural model of noisy strategic choice, and computing one is solved: `pygambit` [15] ships the only production logit-QRE homotopy in any language. `strataq` does not compete on that. It exists because four things researchers need *after* an equilibrium have no implementation anywhere, and are therefore hand-rolled, differently, in every paper that needs them.

1. Nobody implements the Candogan decomposition. Fifteen years after Candogan et al. [8] gave the orthogonal decomposition of a finite game into potential, harmonic and non-strategic flows, there is **no package on any index that implements it**; the public artifacts are one-off notebooks, and the recent theory papers that use it as their baseline coordinate system release no code. The harmonic fraction α is the natural independent variable for “how far is this game from a potential game”, and it has been unavailable in practice. `strataq` computes it with a separable Kronecker transform, near-linear in the payoff tensor size and verified against the projection identities at machine precision.

2. No solver exposes the equilibrium derivative. What a practitioner wants is $d\sigma^*/dh$: how the equilibrium moves when a payoff moves. Every QRE package returns the equilibrium and stops, so comparative statics are done by finite-differencing a re-solved equilibrium — slow, noisy, and *silently wrong across a turning point*, where the re-solve can land on a different branch. The derivative is available in closed form as the strategic resolvent $\chi^{\text{eq}} = (I - SB)^{-1}S$ on the tangent space, and `strataq` returns it directly. The same operator is the uniqueness certificate ($\rho(SB) < 1$), the bifurcation detector, and the kernel of the custom JAX VJP: one factorisation serves comparative statics, criticality warnings and gradients through the solve.

3. Irreversibility estimators cannot see agents. The live packages for time-series irreversibility take a scalar series and return a verdict; they have no concept of an agent, an action or a profile, so they cannot answer the only version of the question a game theorist or a regulator can act on — which player’s response asymmetry produces the entropy production — and the canonical review of coarse-grained bounds [26] ships no code at all. `strataq` constructs the joint-profile Glauber generator implied by logit revision and computes its stationary distribution, currents and exact Schnakenberg entropy production [21], with trajectory estimators (k -block KLD [22], debiased finite-time thermodynamic-uncertainty bound [23, 24]) validated against that exact meter.

4. The estimation workflow has a manual instead of an API. The maintainer of the reference solver co-authored Bland and Turocy [4], “the missing manual” for QRE estimation. A manual exists because the workflow is not in the software: `pygambit.qre.logit_estimate` returns the estimate, the profile and the log-likelihood — no standard error, no interval, no likelihood-ratio test — and its *input* is a profile of aggregated counts, so subject, round and treatment are destroyed before the likelihood is written. Every published interval on λ is hand-rolled, differently in each lab, which makes them incomparable across papers. `strataq`’s `fit()` takes tidy data and keeps the panel: with a `subject` column the bootstrap resamples *subjects*, so within-subject correlation widens the interval instead of being averaged away, and without one it

says so. It prints $\hat{\lambda}$, an interval with its method named, n , the log-likelihood, and likelihood-ratio tests against both nested boundaries, Nash ($\lambda \rightarrow \infty$) and uniform ($\lambda = 0$).

Positioning, stated once. *pygambit computes equilibria; strataq tells you what kind of system you are looking at.* We do not compete on correctness of the QRE correspondence — **pygambit** is our oracle, which converts the incumbent from a competitor into a certificate. **nashpy** [16] is two-player with no QRE; **OpenSpiel** [17] has magnetic mirror descent [18] — one entropy-regularised solution at one fixed regularisation, two-player zero-sum sequential — and no QRE entry in its algorithm table; **QuantEcon** ships the Glauber chain and never closes the loop to a logit fixed point. One practical fact decides adoption: every **pygambit** release to date ships Windows wheels and an sdist only, so Linux, macOS and every Colab notebook compile a C++17 codebase to use the reference solver. **strataq** is pure Python on JAX and installs everywhere with no compiler.

2 Design

Two engines, and a rule about what may cross between them. **strataq** is a uv-workspace monorepo in JAX. Engine 1 is the finite strategic form (dense payoff tensors, normalisation, the Hodge decomposition, the response instruments, the solvers, the exact dynamics); Engine 2 is population games (routing networks, Fisk stochastic user equilibrium [12], population-level susceptibility). A third engine for Bayesian games with type spaces is *deliberately not started*, by a recorded decision, because the finite-strategic-form machinery does not apply to it as written.

Domains — congestion, Blotto, pricing, electricity — are plugins declaring exactly five objects (payoff oracle, action-grid builder, conjugate-field specification, dataset loader, learn page) and making *zero* core changes, enforced by a commit hook and in CI: a domain may import its own declared engine and nothing else, and a commit touching both a domain and the core must cite a recorded architecture decision by id. The invariant is that *a domain that needs core changes is not a domain, it is an engine*.

Refusal is a return value. The conjugate-field declaration names the observable payoff perturbation the response instruments differentiate with respect to — link tolls, budgets, wholesale cost. A domain that cannot name one declares **NONE**, and the response instruments then *refuse to run* rather than returning a number with no operational meaning. The pattern recurs: **fit()** reports a flat likelihood as the interval it truly is (the whole search window) and withholds the point estimate; **diagnose()** reports an unidentifiable coordinate as a bound, or as a named pair of quadrants it cannot separate, never as **None**.

Float64 and the tangent space are correctness requirements, not preferences. JAX x64 is enabled at import: $(I - SB)$ is near-singular exactly where the interesting behaviour is, and float32 is not acceptable there. All linear algebra involving $(I - SB)$ runs on $T = \bigoplus_i \{v : \mathbf{1}^\top v = 0\}$ through an explicit Helmert basis, because $C_i = \text{diag}(\sigma_i) - \sigma_i \sigma_i^\top$ is rank-deficient by construction and skipping the projection manufactures a spurious zero eigenvalue — a false criticality reading. Decomposition always acts on the *normalised* game, since full externality symmetry is sufficient but not necessary for potentiality of the raw game [11].

No literal constants in library code. Tolerances, seeds, grids and λ ranges come from a typed configuration schema, on a named ladder: 10^{-12} for exact identities, 10^{-10} for decomposition and solver convergence, 10^{-8} for oracle cross-checks against an external implementation, 10^{-6} for finite differences. Every stochastic call site takes an explicit PRNG key folded from a root recorded in the resolved configuration; there is no global RNG. Every reading reports both λ and the normalised precision $\bar{\lambda} = \lambda \cdot (\text{payoff range})$, because payoffs are rescaled internally and the scale folds into λ — a confound that has caught one of this project’s own experiments and is now reported by construction.

Two surfaces, one machinery. The upper surface is `strataq.problems` (state a problem, get a solution object); the lower is the instrument layer (`logit_qre`, `chi_equilibrium`, `reciprocity_defect`, `alpha`, `diagnose`). The upper calls the lower and never reimplements it, and the facade’s numbers are gated against the research artifacts they claim to reproduce.

3 What the library computes

A map of the public surface as it stands. Notation: players i with finite action sets, mixed profiles σ_i , precision λ_i ; $C_i = \text{diag}(\sigma_i) - \sigma_i \sigma_i^\top$; $S = \text{blockdiag}(\lambda_i C_i)$; $B_{ij}(a, b) = \partial U_i(a) / \partial \sigma_j(b)$.

Solve. `logit_qre` is a JIT-compiled damped Krasnoselskii–Mann iteration on the logit fixed point. `mirror` adds magnetic mirror descent [18] — a structurally different, last-iterate-convergent method — so that solver dependence can be *tested* rather than assumed away. `homotopy` is a pseudo-arclength branch tracer with fold detection by λ -direction reversal and $\rho(SB)$ recorded along the branch, so a turning point of the correspondence is a labelled event rather than a silent jump. `implicit` differentiates through the fixed point with a custom VJP on the transposed tangent system, sharing the $(I - SB)$ factorisation.

Respond and decompose. `chi_partial` is the opponents-frozen identity $\chi^{\text{part}} = \lambda_i C_i$; `chi_equilibrium` is $\chi^{\text{eq}} = (I - SB)^{-1} S$; `reciprocity_defect` is $\mathcal{R} = \|\chi^{\text{eq}} - \chi^{\text{eq}\top}\|_F / \|\chi^{\text{eq}} + \chi^{\text{eq}\top}\|_F$, zero if and only if the normalised game is potential. Two properties are documented at the call site because both have been misread in print, including by us: $\mathcal{R}$ is a ratio of norms rather than a fraction, so it is unbounded above (matching pennies reads 1.20), and only its *zero* is λ -free — the magnitude scales with λ , so any level carries the λ it was read at. `strategic_spectrum` and `critical_lambda` locate criticality from the same operator, warning rather than returning a huge χ^{eq} near it. `hodge_decompose` and `alpha` give the geometry, and `make_family` constructs games at an exact target α , which turns an α -sweep into a controlled experiment.

Dynamics and thermodynamics. `core.dynamics` builds the Glauber generator on the joint profile space with its stationary distribution, currents J^* , exact σ_{EP} and a uniformised trajectory sampler. `thermo.estimators` holds the data-side estimators; `thermo.nulls` the surrogate classes, including a reversibilized-Markov null matching an observed chain’s persistence, which is the sharp test for pair-level detailed balance; `thermo.protocols` the Hatano–Sasa [25] housekeeping/excess split and stepwise λ -quench protocols with their fluctuation theorems; and `thermo.hs_estimator` the plug-in quench estimator with a standalone `relaxation_gate` that refuses windows which have not settled.

Estimate. `estimate.lam` runs four structurally different λ estimators — frequency MLE with a profile-likelihood interval, autodiff MLE through an unrolled solve, χ moment-matching, dispersion inversion — with an agreement protocol treating their disagreement as a misspecification diagnostic. `estimate.bayes` adds a grid posterior with a self-diagnosing resolution guard, mixture Bayes factors and an expected-free-energy experiment-selection loop. `fit()` packages the workflow of Section 1 for tidy experimental data.

Repeated, evolutionary, extensive. `repeated` writes strategies as Moore machines, so grim trigger and tit-for-tat are data rather than code paths; it answers the folk theorem’s two questions (sustainable at δ ? critical δ ?) in closed form for grim trigger and generically by one-shot deviation, adds a logit analogue in which sustainability is a probability, and measures Edgeworth cycles [28] from alternating logit best response on a price ladder. `evolutionary` has replicator dynamics in continuous and discrete time, rest points by support enumeration classified by tangent-space eigenvalues, the logit dynamic whose rest points are the logit QRE, and the finite-population Moran side; the two meet at one scalar, because the selection intensity β of the Fermi rule *is* the logit precision λ , and `compare_intensity` reports both readings with the gap between them. `extensive` stores trees as flat preorder arrays with a nested-dict constructor that doubles as the JSON format, and implements the two tree passes, the reduced normal form, Kuhn’s theorem [27], backward induction, a catalogue of classics, and the agent QRE [2] — logit at every information set — on the same damped fixed point as the strategic-form solver, cross-validated against `pygambit` (Section 6).

Problems, verdicts, figures. `strataq.problems` exposes eight problem types — pricing, auction, routing, allocation, electricity, repeated, evolutionary, extensive — on one contract: construct, solve, summarise. Problems validate on construction with actionable errors; solutions are frozen dataclasses whose `.diagnostics` carry the response, decomposition and dissipation readings, computed only if asked for. `solve_situation()` assembles a whole recommendation once, including a sensitivity block that re-solves across a ladder of precisions and reports what fraction of it agrees and where the answer first changes — a recommendation holding from $\lambda/4$ to 4λ is a different object from one holding only at the λ you picked, and the API says which one you have. `diagnose()` is the one-call entry point: a verdict, with `explain()` for every warning, band, null and seed, `snippet()` for a runnable reproduction of exactly that reading, and `plot()` for the point in the $(\mathcal{R}, \sigma_{EP})$ plane. `strataq.viz` carries one palette shared by library, application and papers.

4 Calibration: the argument

A meter is only as credible as the systems on which its reading is known in advance. Table 1 is this paper’s argument, and everything else is scaffolding for reading it: fourteen readings on systems whose correct answer is fixed either by an analytic identity or by an external oracle, each with the requirement it had to meet, the number measured, and the committed artifact it regenerates into. Every row runs from a fixed seed under `make reproduce`; every artifact is a JSON record carrying metrics, effect sizes with intervals, n , an n -justification, the seed, the configuration reference and the library version.

Reading	Requirement	Measured	Artifact
$\nabla\psi = \lambda\sigma$, $\nabla^2\psi = \lambda^2C$ (log-partition is the CGF)	10^{-12}	holds	test suite
Gibbs $\pi \propto e^{\lambda\Phi}$ on congestion, $N=2, 3$	10^{-10}	4.7×10^{-16}	<code>gibbs_agreement</code>
σ_{EP} and $ J^* $ on exact potential games	$< 10^{-12}$	4.2×10^{-30} , 9.7×10^{-17}	<code>equilibrium_reads_zero</code>
$\mathcal{R}$ on 5 exact potential games	$< 10^{-10}$	$\leq 8.9 \times 10^{-17}$	<code>reciprocity_potential</code>
$\mathcal{R}$ on RPS-3 / RPS-5 / matching pennies	> 0.1	0.693 / 0.435 / 1.200	<code>reciprocity_harmonic</code>
χ^{eq} vs central finite differences, 50 games	10^{-6}	1.3×10^{-8}	<code>chi_fd_agreement</code>
Implicit-diff VJP vs the resolvent, 20 games	10^{-8}	5.6×10^{-17}	<code>implicit_chi_agreement</code>
Mirror descent vs damped solver, same fixed point	10^{-9}	6.9×10^{-13}	<code>solver_cross_agreement</code>
Real spectrum of SB on potential games, $\lambda \leq 10$	rel. imag $< 10^{-8}$	0 (harmonic: ≥ 0.29)	<code>spectrum_reality</code>
<code>pygambit</code> fixed- λ agreement, 32 games $\times 2 \lambda$	10^{-8}	1.2×10^{-9}	<code>gambit_agreement</code>
Branch tracer through fixed- λ points	10^{-7}	8.6×10^{-13}	<code>branch_agreement</code>
Engine 2: $\nabla\text{Beckmann} = \text{route costs (autodiff); toll } \chi \text{ vs FD}$	$10^{-12}; 10^{-6}$	0; 2.8×10^{-9}	<code>population_identities</code>
Sioux Falls (real TNTF data): $\mathcal{R}$ / DF defect / Fisk KKT	10^{-10} / 10^{-12} / 10^{-8}	5.6×10^{-17} / 0 / 7.1×10^{-15}	<code>sioux_falls_calibration</code>
Blotto: α , $\mathcal{R}$, degenerate constant-payoff null	see text	0.694, 0.118, 5.4×10^{-32}	<code>blotto_readings</code>

Table 1: Calibration summary. Every row regenerates from seed 20260808 via `make reproduce`; artifacts are JSON records under `benchmarks/results/` (Appendix A). “Requirement” is the tolerance declared in the configuration schema before the check ran, not a threshold chosen afterwards to fit the number.

The $\alpha = 0$ anchor: congestion on real network data. Rosenthal congestion games [13] are exact potential games, and in the population form logit route choice *is* Fisk’s stochastic user equilibrium [12], whose convex potential — the Beckmann integral plus scaled entropy — is known analytically. On the Braess diamond the autodiff gradient of the Beckmann integral equals the route-cost vector exactly, which is the project’s sharpest end-to-end code check because it composes the payoff oracle, the network, the potential and the differentiation in one identity. On real Sioux Falls data (76 links, top-8 OD pairs, $k=3$ route sets from the TNTF repository [29]) the Fisk KKT conditions hold to 7.1×10^{-15} , DF is symmetric to exact zero, the toll-response reciprocity defect reads $\mathcal{R} = 5.6 \times 10^{-17}$, and the toll susceptibility matches finite differences to 2.8×10^{-9} with the sign verified against actual re-solves. The scope is stated once and not softened: route sets are k -shortest restrictions and the OD set is a subset, so computed link flows are *not* comparable to the repository’s best-known equilibrium flows (a diagnostic gap of 0.735 is reported as a diagnostic, never a certificate); the instrument identities are exact within the route set regardless. One practical fact is documented rather than hidden: the damped logit-assignment map is *not* a contraction on steep-BPR networks — it two-cycles on the Braess diamond — which is why the SUE path is damped Newton on the convex program.

The $\alpha > 0$ anchor: Colonel Blotto. Two-player Blotto over lattice-simplex allocation grids, with budgets as an exactly observable conjugate field. The symmetric budget-3, three-field instance reads $\alpha = 0.694$ and $\mathcal{R} = 0.118$ at $\lambda = 1.5$; an asymmetric instance reads $\sigma_{\text{EP}} = 0.098$ with visible circulation; a degenerate instance (equal field values, budget 2, every profile ties) has constant payoffs and every meter reads exactly zero ($\sigma_{\text{EP}} = 5.4 \times 10^{-32}$) — a correct null found while writing the positive test and kept as a regression. One calibration surprise corrected a documented assumption: Blotto is *mixed*, not pure harmonic ($\alpha \approx 0.7$, not 1). Zero-sum does not imply harmonic-pure; RPS ($\alpha = 1$ exactly) remains the pure extreme, so the right-hand anchor of the bracket sits at ≈ 0.7 for a realistic strategic system.

The dissipation axis is observable, and the bound is the reportable number. Against the exact Schnakenberg meter across a ten-level α sweep, the k -block KLD estimator [22] attains Spearman rank correlation 1.0 with per-level relative error of order 1%, and the debiased finite-time thermodynamic-uncertainty bound [23, 24] lies below the exact value at every level with median tightness 0.850 (`estimator_alpha_sweep`). Two implementation facts are recorded because they were learned the hard way and change the number: the estimator must be debiased for $E[\bar{J}^2]$ and for Jensen on $1/\text{Var}$, and windows must be fixed-*horizon* — fixed-jump-count windows overshoot by about $1.5\times$. On the named ground-truth games the estimator reads 3×10^{-5} where the exact answer is zero, and 2.228 against an exact 2.239 on RPS-3 (`estimator_ground_truth`).

5 Validation against reference implementations

Four of the calibration rows deserve their own paragraph, because in them an external implementation or a theorem someone else proved — not our own algebra — decides whether we are right.

Against `pygambit`, strategic form. `pygambit` [15] implements the logit tracing homotopy of Turocy [14]. On 32 seeded games at two precisions each, our fixed- λ profiles agree with it to a maximum profile gap of 1.2×10^{-9} against a declared oracle tolerance of 10^{-8} (`gambit_agreement`). The branch tracer is checked the same way but along the correspondence rather than at a point, reproducing the direct solve at fixed- λ crossings to 8.6×10^{-13} (`branch_agreement`). `pygambit` is an optional test-time oracle, never a runtime requirement.

Against `pygambit`, extensive form. The agent QRE is cross-validated against `pygambit`’s extensive-form logit solver on four classic trees — entry deterrence at $\lambda \in \{0.5, 2.0, 5.0\}$, the four-stage centipede, Selten’s horse and Kuhn poker — each required to sit inside the 10^{-8} oracle tolerance in maximum behaviour-strategy gap. This check lives in the test suite rather than in a committed artifact, and it is `skipif`-ed when `pygambit` is absent, so a CI run without the optional dependency passes a suite whose oracle checks never executed. That is a real gap in the evidence and it is named again in Section 8.

Against finite differences, and against a second derivative path. That $\chi^{\text{eq}} = (I - SB)^{-1}S$ is the equilibrium’s derivative is checked two independent ways. Central finite differences of the solved equilibrium under a payoff perturbation agree with the closed form to 1.3×10^{-8} across 50 random games at a declared finite-difference tolerance of 10^{-6} (`chi_fd_agreement`);

the residual is truncation, not solver error. Independently, the implicit-differentiation VJP — which solves the transposed tangent system instead of forming the resolvent — reproduces the same matrix to 5.6×10^{-17} on 20 games (`implicit_chi_agreement`), which is the statement that the two code paths are the same mathematics rather than merely close.

Against an analytically known potential. The population engine’s identities are exact rather than approximate: the autodiff gradient of the Beckmann integral equals the route-cost vector to exact zero, the DF symmetry defect is exact zero, and the Fisk KKT spread is 7.1×10^{-15} on real network data. A library whose central instrument is a potentiality test should read zero where a theorem says zero, and that is where it is checked.

6 Reproducibility, and one instructive failure

A number is not a result here until it has walked a fixed path. Criteria are written into `config/experiments/<unit>.yaml` and *committed before the run*, with the commit verified landed by `git log` — a rule that exists because a pre-registration once aborted on a hook failure and a review caught the configuration staged but uncommitted while results already existed. The experiment then runs, writing a resolved configuration snapshot beside its artifact; the artifact lands in `benchmarks/results/` with metrics, effect sizes and intervals, n , an n -justification, the seed and the configuration reference. A *gate* (`gates/<unit>.yaml`) checks five dimensions — code (tests, strict types, lint, coverage, no stubs), domain (the artifact exists, reports **passed**, carries a seed), statistical (effect sizes with intervals, justified n , recorded seeds), documentation, and adversarial (a red-team signoff plus a disposition for *every* objection) — and nothing marks a unit green except the deterministic gate runner observing those checks pass. Anti-gaming rules are mechanical: deleting or `xfail`-ing a failing test blocks the commit, weakening a domain gate requires a recorded architecture decision, and gate regressions fail CI. Where a gate was written after the measurement as a regression contract rather than as a prediction, the gate file says so at the top; where a unit is not signed off, the gate is **red on the board** rather than absent from it.

One instructive failure (F-0018). *Twenty-three gates were green while every solver call in the shipped wheel was broken.* The packaged `strataq/core/base.yaml` fallback that `base_config()` documents was never included in the distribution, so from an installed wheel every solver-touching entry point raised `FileNotFoundError` — the toolkit’s thermodynamics and rationality paths were dead for anyone outside the repository, and only the pure-`numpy` paths worked. Separately, `__version__` was a hand-edited literal that had drifted from `pyproject.toml`. Every one of those 23 gates was correct; nothing gated the packaging boundary, so the whole suite was true in-repo and false for outsiders.

Both defects were caught by a pre-upload smoke test and fixed before publication. Four release-integrity tests exist because of it — the config file is in the wheel, is byte-identical to the repository’s, loads standalone through the schema, and `__version__` matches `pyproject.toml` — together with a CI wheel smoke test that now exercises a *solver* path rather than the `numpy`-only entry point the bug had hidden behind, and a standing rule: no release without an out-of-tree smoke of every public entry point. Installation from the public index into a clean virtual environment reproduces the committed empirical reading it is checked against.

We report this at length because it is the most useful paragraph in the paper for another maintainer. A suite tells you only that the suite passed; before trusting one, ask what it would have to *see* in order to fail, and whether it could see it. Ours could not see the wheel — which is also why the skippable extensive-form oracle check is in Section 8 rather than implied by a green badge.

Reproducing. `uv sync -all-packages` then `make reproduce` regenerates every artifact in Table 1 from its recorded seed; `uv run python gates/run_gates.py -check` re-runs every gate. Appendix A maps each number in this paper to the artifact it came from.

7 Limitations and known gaps

The exact dynamics layer is dense and small. The Glauber generator is built over $\prod_i m_i$ joint profiles as a dense matrix. That is a cost limit, not a correctness limit, but it is a hard one: the exact entropy production, currents and stationary distribution are intended for small games, and the deployed API refuses dense-dynamics requests above a few hundred joint states. Larger systems must go through the trajectory estimators, where the reportable statement is the certified lower bound rather than the point estimate.

The product surface is tested but not gated. The instrument layer — solvers, response, decomposition, dynamics, estimators — is gated unit by unit. The surface built on top of it during the most recent development phase (`problems`, `solve_situation`, `fit`, `diagnose`, `viz`, `repeated`, `evolutionary`, `extensive`) has tests but no gate, no acceptance artifact and no claims ledger entry. That is recorded as an architecture decision with a named gate and a date for each item, rather than left to be inferred from a green badge. Until those close, the honest statement is: *the instrument layer is gated; the product surface on top of it is tested but not gated.*

The extensive-form oracle checks are skippable. The four `pygambit` cross-validations of the agent QRE are marked `skipif(not HAVE_GAMBIT)`. Since `pygambit` is an optional dependency, an environment without it runs the extensive-form suite with its only external check silently absent. The fix is to make the oracle comparison a committed benchmark artifact — as the strategic-form comparison already is — so it appears in the record rather than in a test that may not have run.

The public surface is broader than `__all__`, and `golden/` is empty. Several modules export more than their documented interface, so users can reach objects whose stability is not promised. There are no cross-version golden files, so numerical drift between releases would be caught by the calibration artifacts (which are regenerated) rather than by a frozen reference (which is not).

Scope of the readings, not of the code. Two limits are worth stating plainly because users will otherwise infer more than the artifacts support. First, $\mathcal{R}$ ’s magnitude is λ -dependent, so cross- λ comparison of levels is not licensed; only the zero test is λ -free. Second, the scientific results this library was built to obtain — reported separately [30] — carry their own scope, in

particular that the α -stratified co-movement design behind them is a *two-player* instrument, and the library’s documentation carries that restriction wherever it quotes those numbers.

Things we deliberately did not build. A Bayesian-games engine with type spaces is deferred by a recorded decision rather than half-implemented. There is no performance leaderboard, because the comparison that matters is agreement with the oracle. And there is no claim that λ estimated from field data measures rationality: it absorbs unmodelled heterogeneity and misspecified payoffs [5], which is why the estimation surface reports four estimators, their disagreement, and a refusal where the likelihood is flat.

Availability

`strataq` is Apache-2.0 and on PyPI (`pip install strataq`); the repository (<https://github.com/SharathSPhD/sage>) carries the library, the experiments, the gate records, the claim ledger and the findings log. Dominick’s-derived artifacts are CC-BY-NC-4.0 and ride with the data, never with the package.

A Every number’s provenance

Every quantitative statement in this paper maps to a machine-readable artifact in `benchmarks/results/` (JSON, schema `strataq_bench.BenchmarkResult`: metrics, effect sizes with intervals, n , n -justification, seed, configuration reference, library version, timestamp). All regenerate with `make reproduce` (seed 20260808 unless stated). The gate records in `gates/*.yaml` carry, for each unit, the claim, the section-by-section pass state, and the adversarial reviewer’s objections with their dispositions.

Statement in text	Artifact
Calibration table (Table 1)	the artifacts named in its rows
χ^{eq} vs finite differences, 1.3×10^{-8}	<code>chi_fd_agreement.json</code>
Implicit-diff VJP vs the resolvent, 5.6×10^{-17}	<code>implicit_chi_agreement.json</code>
<code>pygambit</code> fixed- λ agreement, 1.2×10^{-9}	<code>gambit_agreement.json</code>
Branch tracer agreement, 8.6×10^{-13}	<code>branch_agreement.json</code>
Mirror descent vs damped solver, 6.9×10^{-13}	<code>solver_cross_agreement.json</code>
Sioux Falls: $\mathcal{R}$, DF defect, Fisk KKT, UE diagnostic gap	<code>sioux_falls_calibration.json</code>
Engine-2 identities (Beckmann gradient, KKT, DF , toll- χ)	<code>population_identities.json</code>
Blotto readings incl. the degenerate null	<code>blotto_readings.json</code>
$\mathcal{R}$ on potential / harmonic games	<code>reciprocity_potential.json</code> , <code>reciprocity_harmonic.json</code>
Gibbs stationarity; σ_{EP} and J^* zeros; SB spectrum	<code>gibbs_agreement.json</code> , <code>equilibrium_reads_zero.json</code> , <code>spectrum_reality.json</code>
KLD rank correlation 1.0; TUR median tightness 0.850	<code>estimator_alpha_sweep.json</code> , <code>estimator_ground_truth.json</code>
Toolkit facade reproduces its research numbers	<code>toolkit_verdicts.json</code>
AQRE vs <code>pygambit</code> extensive-form logit	test suite (<code>tests/extensive/</code>), <i>not</i> a committed artifact — see Section 8

Two pointers, and a correction to an earlier draft of this appendix. The findings log (`memory/findings.md`) records each anomaly with the configuration that produced it, its chase

status and its resolution, including retractions — an earlier version of this appendix stated that the log covered entries **F-0001–F-0007**, which was true of a draft written in August 2026 and is now stale by seventeen entries. The log currently runs **F-0001–F-0024**, and the entries a reader of *this* paper should look at are F-0002 (the “ $\mathcal{R}$ is λ -free” claim corrected to apply only to the zero test), F-0005 (Blotto is mixed, not pure harmonic), F-0011 (the first empirical $\mathcal{R}$), and F-0018 (the release-integrity failure of Section 7). The claim ledger (`memory/claims.md`) carries every scientific claim’s confidence tier and its history, including downgrades; the architecture decisions (`memory/decisions.md`) carry every divergence from the plan, including the two units currently on the main branch whose gates are red.

References

- [1] McKelvey, R.D., Palfrey, T.R.: Quantal response equilibria for normal form games. *Games and Economic Behavior* 10(1), 6–38 (1995).
- [2] McKelvey, R.D., Palfrey, T.R.: Quantal response equilibria for extensive form games. *Experimental Economics* 1, 9–41 (1998).
- [3] Goeree, J.K., Holt, C.A., Palfrey, T.R.: *Quantal Response Equilibrium: A Stochastic Theory of Games*. Princeton (2016).
- [4] Bland, J.R., Turocy, T.L.: Quantal response equilibrium as a structural model for estimation: the missing manual. *Games and Economic Behavior* (2025).
- [5] Haile, P.A., Hortagsu, A., Kosenok, G.: On the empirical content of quantal response equilibrium. *American Economic Review* 98(1), 180–200 (2008).
- [6] Monderer, D., Shapley, L.S.: Potential games. *Games and Economic Behavior* 14(1), 124–143 (1996).
- [7] Blume, L.E.: The statistical mechanics of strategic interaction. *Games and Economic Behavior* 5(3), 387–424 (1993).
- [8] Candogan, O., Menache, I., Ozdaglar, A., Parrilo, P.A.: Flows and decompositions of games: harmonic and potential games. *Mathematics of Operations Research* 36(3), 474–503 (2011).
- [9] Sandholm, W.H.: *Population Games and Evolutionary Dynamics*. MIT Press (2010).
- [10] Balduzzi, D., et al.: The mechanics of n -player differentiable games. *ICML* (2018).
- [11] Legacci, D., Mertikopoulos, P., Pradelski, B.S.A.: A geometric decomposition of finite games: convergence vs. recurrence under exponential weights. *ICML* (2024). arXiv:2405.07224.
- [12] Fisk, C.: Some developments in equilibrium traffic assignment. *Transportation Research B* 14(3), 243–255 (1980).
- [13] Rosenthal, R.W.: A class of games possessing pure-strategy Nash equilibria. *Int. J. Game Theory* 2, 65–67 (1973).

- [14] Turocy, T.L.: A dynamic homotopy interpretation of the logistic quantal response equilibrium correspondence. *Games and Economic Behavior* 51(2), 243–263 (2005).
- [15] Savani, R., Turocy, T.L.: Gambit: the package for computation in game theory, version 16.7.0. <https://gambitproject.readthedocs.io> (accessed 12 August 2026).
- [16] Knight, V., Campbell, J.: Nashpy: a Python library for the computation of Nash equilibria. *Journal of Open Source Software* 3(30), 904 (2018).
- [17] Lanctot, M., et al.: OpenSpiel: a framework for reinforcement learning in games. arXiv:1908.09453 (2019).
- [18] Sokota, S., et al.: A unified approach to reinforcement learning, quantal response equilibria, and two-player zero-sum games. *ICLR* (2023).
- [19] Matějka, F., McKay, A.: Rational inattention to discrete choices. *American Economic Review* 105(1), 272–298 (2015).
- [20] Fudenberg, D., Iijima, R., Strzalecki, T.: Stochastic choice and revealed perturbed utility. *Econometrica* 83(6), 2371–2409 (2015).
- [21] Schnakenberg, J.: Network theory of microscopic and macroscopic behavior of master equation systems. *Reviews of Modern Physics* 48, 571–585 (1976).
- [22] Roldán, É., Parrondo, J.M.R.: Estimating dissipation from single stationary trajectories. *Physical Review Letters* 105, 150607 (2010).
- [23] Barato, A.C., Seifert, U.: Thermodynamic uncertainty relation for biomolecular processes. *Physical Review Letters* 114, 158101 (2015).
- [24] Horowitz, J.M., Gingrich, T.R.: Proof of the finite-time thermodynamic uncertainty relation for steady-state currents. *Physical Review E* 96, 020103 (2017).
- [25] Hatano, T., Sasa, S.: Steady-state thermodynamics of Langevin systems. *Physical Review Letters* 86, 3463 (2001).
- [26] Seifert, U.: Universal bounds on entropy production from fluctuating coarse-grained trajectories. *Nature Reviews Physics* 8, 493–507 (2026).
- [27] Kuhn, H.W.: Extensive games and the problem of information. In: *Contributions to the Theory of Games II*, Princeton (1953).
- [28] Maskin, E., Tirole, J.: A theory of dynamic oligopoly II: price competition, kinked demand curves, and Edgeworth cycles. *Econometrica* 56(3), 571–599 (1988).
- [29] Stabler, B., Bar-Gera, H., Sall, E.: Transportation networks for research. github.com/bstabler/TransportationNetworks.
- [30] Sathish, S.: The irreversibility plane: response asymmetry and dissipation are independent coordinates of strategic non-equilibrium. Working draft, 2026. In the same repository.